```
mysql> show tables;
+-----------------------------------------------+
| Tables_in_cybersecurity_incident_management |
+-----------------------------------------------+
| assets                                        |
| evidence                                      |
| incident_assets                               |
| incident_comments                             |
| incident_threats                              |
| incidents                                     |
| response_actions                              |
| security_analysts                             |
| threat_categories                             |
| threats                                       |
| users                                         |
+-----------------------------------------------+
11 rows in set (0.01 sec)
```

```
mysql> SELECT i.incident_id, i.title, i.severity, a.name AS analyst FROM Incidents i JOIN Security_Analysts a ON i.assigned_to = a.analyst_id;
+-------------+-----------------------------------+----------+-----------------+
| incident_id | title                             | severity | analyst         |
+-------------+-----------------------------------+----------+-----------------+
|           1 | Suspicious Phishing Email         | HIGH     | Rahul Patil     |
|           2 | Multiple Failed SSH Login Attempts | HIGH     | Amit Shah       |
|           3 | Ransomware Detected               | CRITICAL | Neha Joshi      |
|           4 | SQL Injection Attempt             | CRITICAL | Amit Shah       |
|           5 | Suspicious Network Traffic        | MEDIUM   | Vikram Deshmukh |
+-------------+-----------------------------------+----------+-----------------+
5 rows in set (0.07 sec)
```

```
mysql> SELECT i.incident_id, i.title, u.name AS reporter, i.severity FROM Incidents i JOIN Users u ON i.reported_by = u.user_id;
+-------------+-----------------------------------+----------------+----------+
| incident_id | title                             | reporter       | severity |
+-------------+-----------------------------------+----------------+----------+
|           1 | Suspicious Phishing Email         | Arjun Patil    | HIGH     |
|           2 | Multiple Failed SSH Login Attempts | Sneha Kulkarni | HIGH     |
|           3 | Ransomware Detected               | Rohan Deshmukh | CRITICAL |
|           4 | SQL Injection Attempt             | Priya Shah     | CRITICAL |
|           5 | Suspicious Network Traffic        | Aditya Joshi   | MEDIUM   |
+-------------+-----------------------------------+----------------+----------+
5 rows in set (0.02 sec)
```

```
mysql> DELIMITER //
mysql> CREATE PROCEDURE Generate_Analyst_Report(IN p_analyst_id INT)
    -> BEGIN
    -> DECLARE done INT DEFAULT 0;
    -> DECLARE v_incident_id INT;
    -> DECLARE v_title VARCHAR(200);
    -> DECLARE v_severity VARCHAR(20);
    -> DECLARE v_status VARCHAR(30);
    -> DECLARE incident_cursor CURSOR FOR
    -> SELECT incident_id, title, severity, status
    -> FROM Incidents
    -> WHERE assigned_to = p_analyst_id;
    -> DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;
    -> OPEN incident_cursor;
    -> report_loop: LOOP
    -> FETCH incident_cursor
    -> INTO v_incident_id, v_title, v_severity, v_status;
    -> IF done = 1 THEN
    -> LEAVE report_loop;
    -> END IF;
    -> SELECT
    -> v_incident_id AS Incident_ID,
    -> v_title AS Title,
    -> v_severity AS Severity,
    -> v_status AS Status;
    -> END LOOP;
    -> CLOSE incident_cursor;
    -> END //
Query OK, 0 rows affected (0.04 sec)

mysql> DELIMITER ;
mysql> CALL Generate_Analyst_Report(1);
+-------------+---------------------------+----------+---------------+
| Incident_ID | Title                     | Severity | Status        |
+-------------+---------------------------+----------+---------------+
|           1 | Suspicious Phishing Email | HIGH     | INVESTIGATING |
+-------------+---------------------------+----------+---------------+
1 row in set (0.02 sec)

Query OK, 0 rows affected (0.02 sec)
```

```
mysql> DELIMITER //
mysql>
mysql> CREATE TRIGGER Incident_Status_Update
    -> BEFORE UPDATE ON Incidents
    -> FOR EACH ROW
    -> BEGIN
    ->     IF NEW.status IN ('RESOLVED', 'CLOSED')
    ->        AND (OLD.status IS NULL
    ->             OR OLD.status NOT IN ('RESOLVED', 'CLOSED')) THEN
    ->
    ->         SET NEW.resolved_at = CURRENT_TIMESTAMP;
    ->
    ->     END IF;
    -> END //
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> DELIMITER ;
mysql> UPDATE Incidents
    -> SET status = 'RESOLVED'
    -> WHERE incident_id = 1;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql> DELIMITER //
mysql> CREATE FUNCTION Calculate_Risk_Score( p_severity VARCHAR(20), p_threat_level VARCHAR(20) )
    -> RETURNS INT
    -> DETERMINISTIC
    -> BEGIN
    -> DECLARE severity_score INT DEFAULT 1;
    -> DECLARE threat_score INT DEFAULT 1;
    -> SET severity_score = CASE UPPER(p_severity)
    -> WHEN 'LOW' THEN 1
    -> WHEN 'MEDIUM' THEN 2
    -> WHEN 'HIGH' THEN 3
    -> WHEN 'CRITICAL' THEN 4
    -> ELSE 1 END;
    -> SET threat_score = CASE UPPER(p_threat_level)
    -> WHEN 'LOW' THEN 1
    -> WHEN 'MEDIUM' THEN 2
    -> WHEN 'HIGH' THEN 3
    -> WHEN 'CRITICAL' THEN 4
    -> ELSE 1
    -> END;
    -> RETURN severity_score + threat_score;
    -> END //
Query OK, 0 rows affected (0.02 sec)

mysql> DELIMITER ;
mysql>  SELECT Calculate_Risk_Score('LOW', 'LOW') AS Risk_Score;
+------------+
| Risk_Score |
+------------+
|          2 |
+------------+
1 row in set (0.00 sec)
```